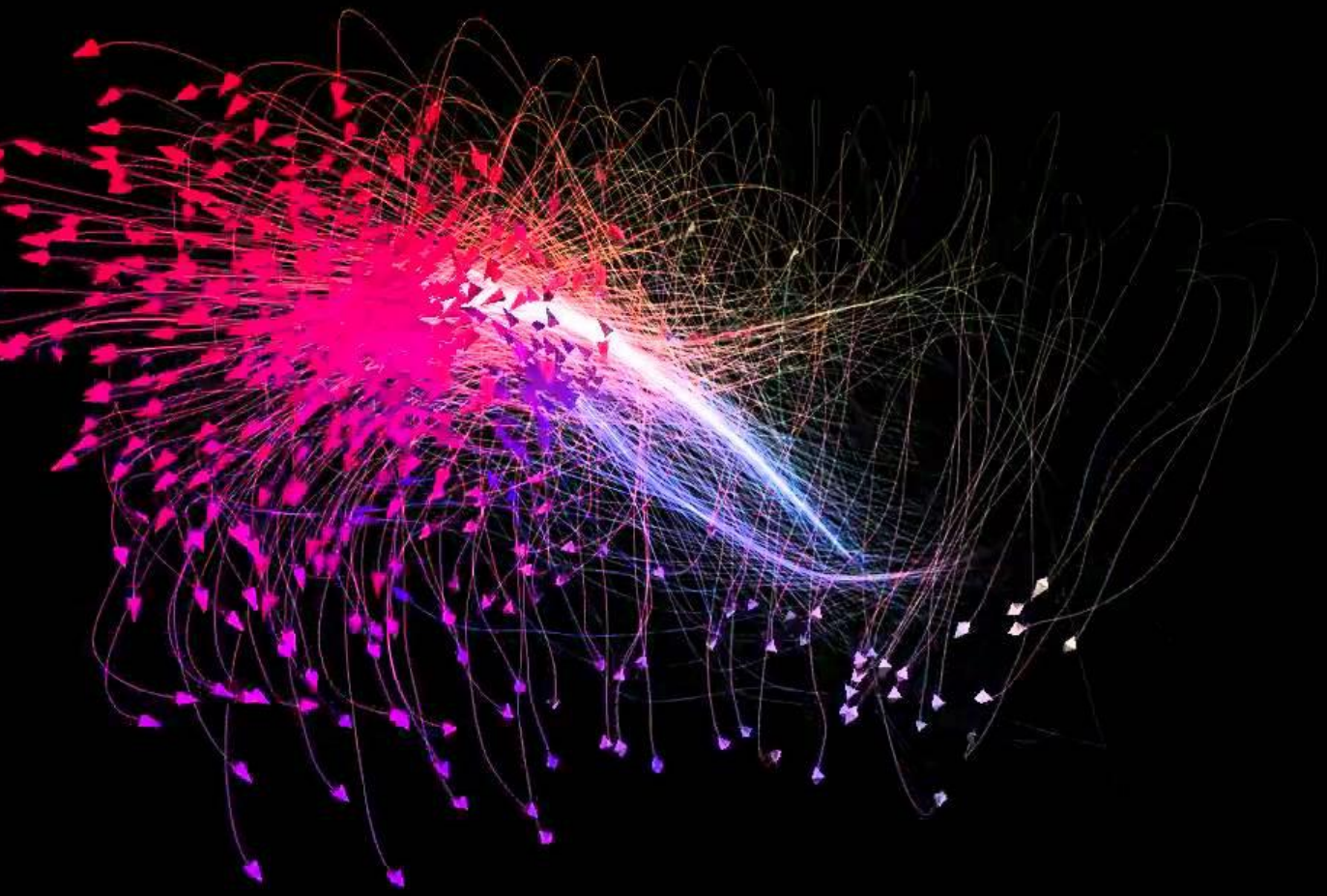
Comparison of Algorithms



Aston University

Outline

- At the end of this lecture you should…
  - …know which types of comparative claims we can reasonably make about the performance of optimisation algorithms,
  - …be able to formulate these claims such that they are testable,
  - …design an experiment to test your claims.



Which algorithm is best?



Which is the best way to travel?











Which is the best way to travel?

- What is the meaning of the word “**best**”?
  - fastest?
  - fastest that fits into my garage?
  - fastest that fits into my garage and which I can afford?
  - most fuel efficient?
  - …
- Means of transport have different properties. Unlikely to find one which is better than another **in all respects.**
- Need to decide which properties are important to us.
- Same goes for optimisation algorithms.



Comparing optimisation algorithms

- When solving an optimisation problem, we are typically interested in:
  - ensuring that **constraints** are satisfied,
  - the **quality** of our best solution,
  - the **resources** available with which to find this solution.
- We will assume that we can find solutions satisfying the constraints.
- We are usually given a means to evaluate **quality** (cost/fitness/objective function)
- **Cost** can be defined in terms of some resource. Often:
  - (CPU) time,
  - number of (cost/fitness/objective) function evaluations.



Comparing optimisation algorithms

- Two algorithms, A and B are to be applied to a problem.
- How would you test the following claims?
- **Claim 1:** After 1000 function evaluations, algorithm A finds a better cost solution than algorithm B.
- **Claim 2:** Algorithm B needs less CPU time to find a solution of cost 5.0 or below than algorithm A does.
- **Claim 3:** Algorithm A finds a lower cost solution than algorithm B in less time.



Comparing optimisation algorithms

- Claims 1 & 2 are easy to think of a test for.
- It is not clear what claim 3 means.
- Experimental approach. Fix either:
  - **Budget** – resources that we can afford
  - **Target value** – quality threshold which we need to meet
- In a real scenario, these will be motivated by real constraints:
  - **Budget** – I need a timetable for the university term before the term starts.
  - **Target value** – my solution must be at least as good as my competitor's.



Which algorithm is fastest?



Which algorithm is fastest?

- If left unqualified, "**best**" is an imprecise term…
- …but we have seen that we can think of a concrete definition for **fastest**
- Algorithm A could be said to **solve a problem faster** than algorithm B if it finds **a solution of some predefined quality in less time**
- Algorithm A could be said to be **faster** than algorithm B if, **given a random problem**, it tends to find **the best possible solution in less time**



Which is faster?



vs



Which is faster?

- Horse vs Human
- Depends on the details of the journey!



Which is faster?



VS



- Distance: 10 miles
- Surface: Concrete Road

Which is faster?





Which is faster?

- Silly examples, but they illustrate an important point:
- If you give me a setting in which one option outperforms another…

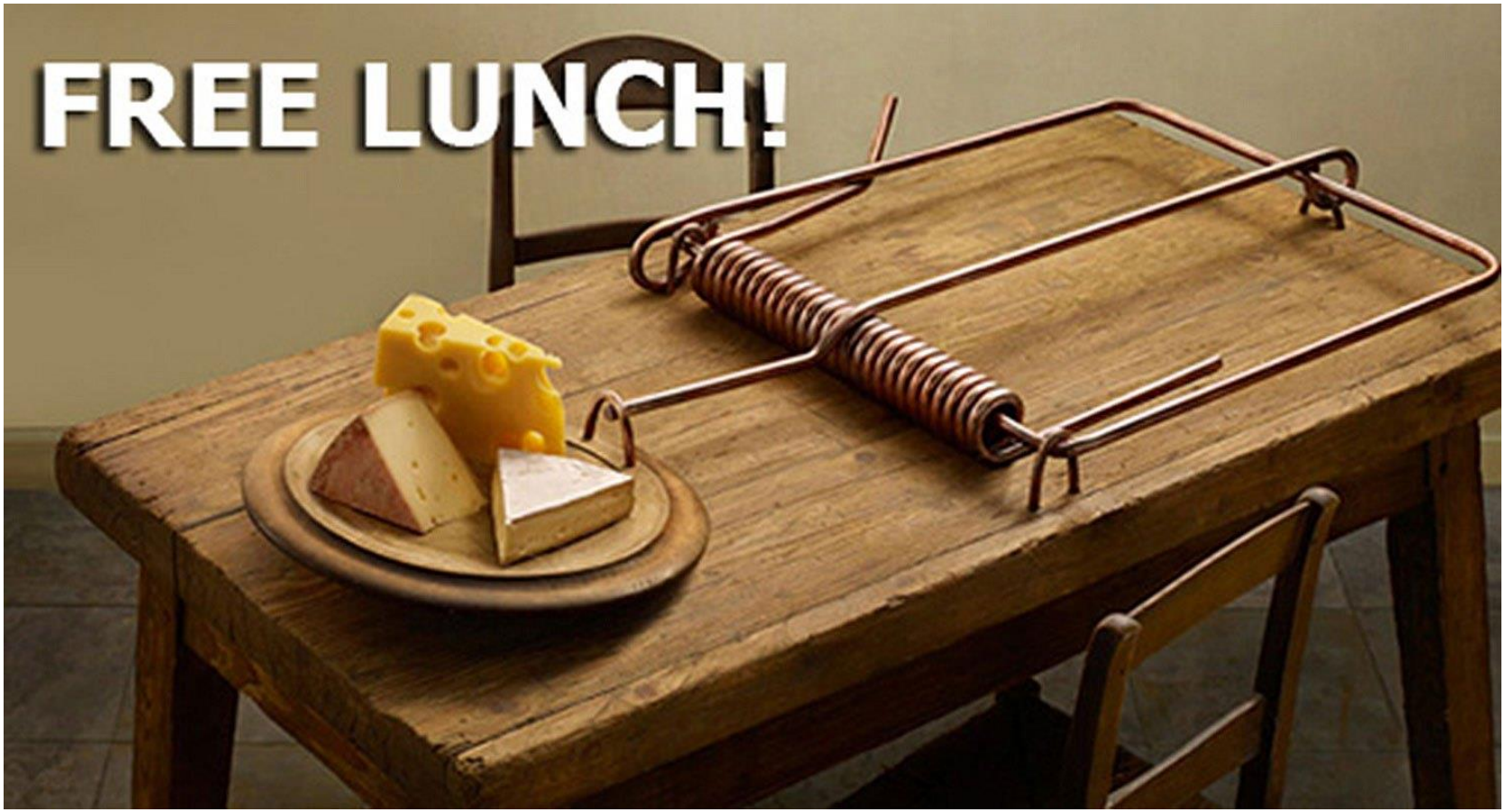




- …then I can modify the setting to reverse the result.






FREE LUNCH!

No Free Lunch Theorem

- A similar argument leads us to the **no free lunch theorem** for optimisation

- The theorem says (more or less) that the cost of solving an optimisation problem, averaged over all possible problems is the same for all algorithms
  - There is **no fastest algorithm**
  - In fact, there is **no algorithm faster than random search**

- Then why have we spent so long teaching you about optimisation algorithms!?



Problem Dependency

- The cost of solving an optimisation problem, averaged over all possible problems is the same for all algorithms.
- The key to unpicking this statement is the "*averaged over all possible problems"* part.
- We might not be able to get good performance for all problems, but we can get **good performance for some class of problems**.



Problem Dependency and Claims

- Which claims can we make?
- **Claim 1:** After 1000 function evaluations, algorithm A finds a shorter tour on TSP instance Ulysses16 than algorithm B does.
- **Claim 2:** On average, after 1000 function evaluations, algorithm A finds shorter TSP tours than algorithm B does.
- **Claim 3:** On average, after 1000 function evaluations, algorithm A finds a better solution to any optimisation problem than algorithm B does.



Problem Dependency and Evidence

- How would I gather **evidence** for a claim like: *"On average, after 1000 function evaluations, algorithm A finds shorter TSP tours than algorithm B does."*
- Test on all possible instances of the TSP?
  - Not possible – an infinite number exist!
- Need a set which I can claim are representative.
- "Representativeness" difficult to argue for for such a broad class – consider reducing the scope of the claim.



How can I test comparative performance claims?



Strength of Claim

- Let's say that I have some fixed metric and problem representative of some class P
  - I try solving the problem using two stochastic algorithms: A and B
  - I observe that A has better performance than B (according to the metric)

- Can I say that A is better than B, according to the metric, on the problem?



Stochasticity

- Which of these two dice will show a higher number next time I roll them?



- Most of the optimisation algorithms we have studied/will study in this course are **stochastic**.
  - They incorporate randomness
  - One observation (one “throw of the dice”) does not tell us everything that we need to know.

Strength of Claim

- Let's say that I have some fixed metric and problem representative of some class P
  - I try solving the problem using two stochastic algorithms: A and B
  - I observe that A has better performance than B (according to the metric)
- Can I say that A is better than B, according to the metric, on the problem?
- Which of the following claims is stronger?
  - Algorithm A ***can*** outperform (by some metric) algorithm B on a problem of type P
  - Algorithm A ***will, on average,*** outperform (given some metric) algorithm B on problems of type P
- Which do I have evidence for?
- A claim that one approach is better than another is strong. We need good evidence to make it.



Strength of Evidence

- A claim that one approach is better than another is strong. We need good evidence to make it.

- What counts as good evidence?
  - A big difference in performance?
  - Lots of observations?

- How big a difference do we need to see?
- How many observations?

- **Statistical hypothesis testing** is a framework which allows us to answer these questions.



What is **statistical hypothesis** testing?



Statistical Hypothesis Testing

- We have some claim that we want to prove (the **alternate hypothesis**)
  - *Algorithm A **is, on average, better** than algorithm B on problem P in terms of metric M.*
- This is placed in opposition to some default position (the **null hypothesis**)
  - *Algorithm A **is, on average, no better** than algorithm B on problem P in terms of metric M.*
- We then check whether the evidence available is sufficient to **falsify the null hypothesis**
  - If it is, then we can accept an *appropriately constructed* alternative hypothesis.



How Can My Conclusions be Wrong?

	Null Hypothesis Rejected	**Null Hypothesis Not Rejected**
Null Hypothesis True	Type I Error	Correct Judgement
Null Hypothesis False	Correct Judgement	Type II Error

- In hypothesis testing, the focus is usually on **type I error**
- Want to have a probability of **type I error** no more than a fixed **significance level** α
- It is quite common to set $\alpha = 0.05$



Is Hypothesis Testing Biased?

- It seems like this setup biases us towards accepting the null hypothesis:
  - If I have lots of evidence ($\alpha = 0.05$), **then I have good reason to accept** the alternative hypothesis
  - Otherwise, **I have good reason to accept** the null hypothesis

- Be careful! These claims are stronger than we can justify.

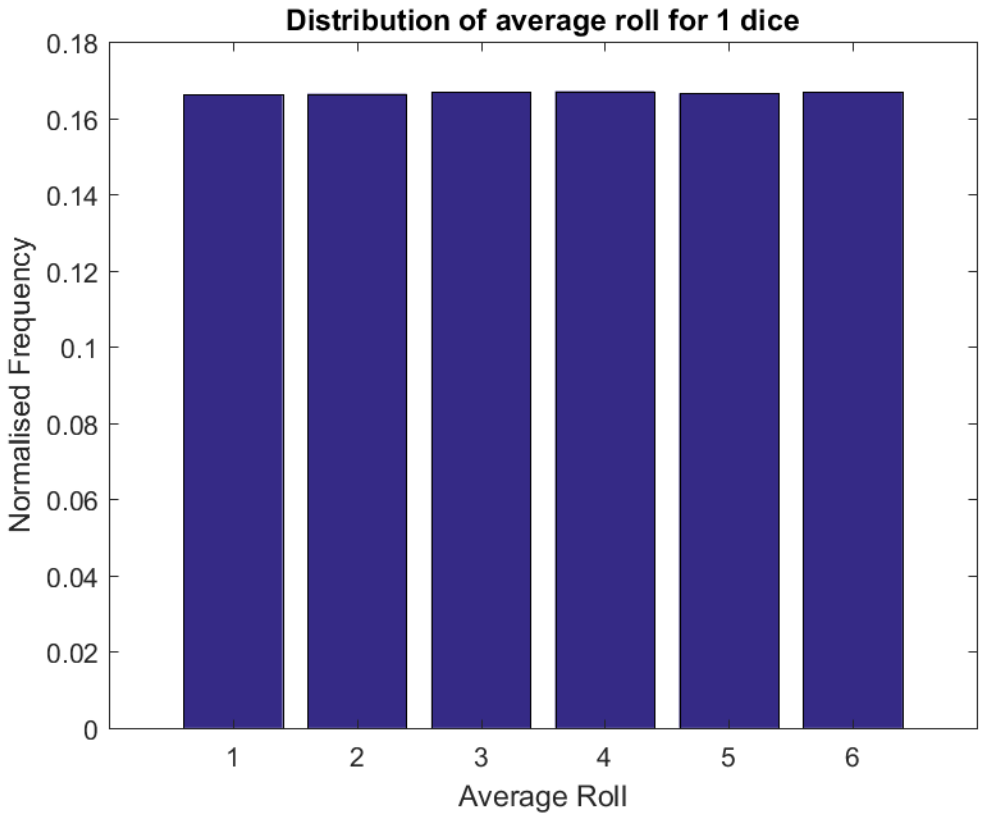
- What we can actually claim:
  - If I have lots of evidence ($\alpha = 0.05$), then **I have good reason to reject** the null hypothesis
    - If the alternative hypothesis is the negation of the null hypothesis, then **I have good reason to accept** the alternative
  - Otherwise, **I have no good reason to reject** the null hypothesis

- Lack of evidence that something is false is not the same as evidence that it is true.

Are My Dice Crooked?



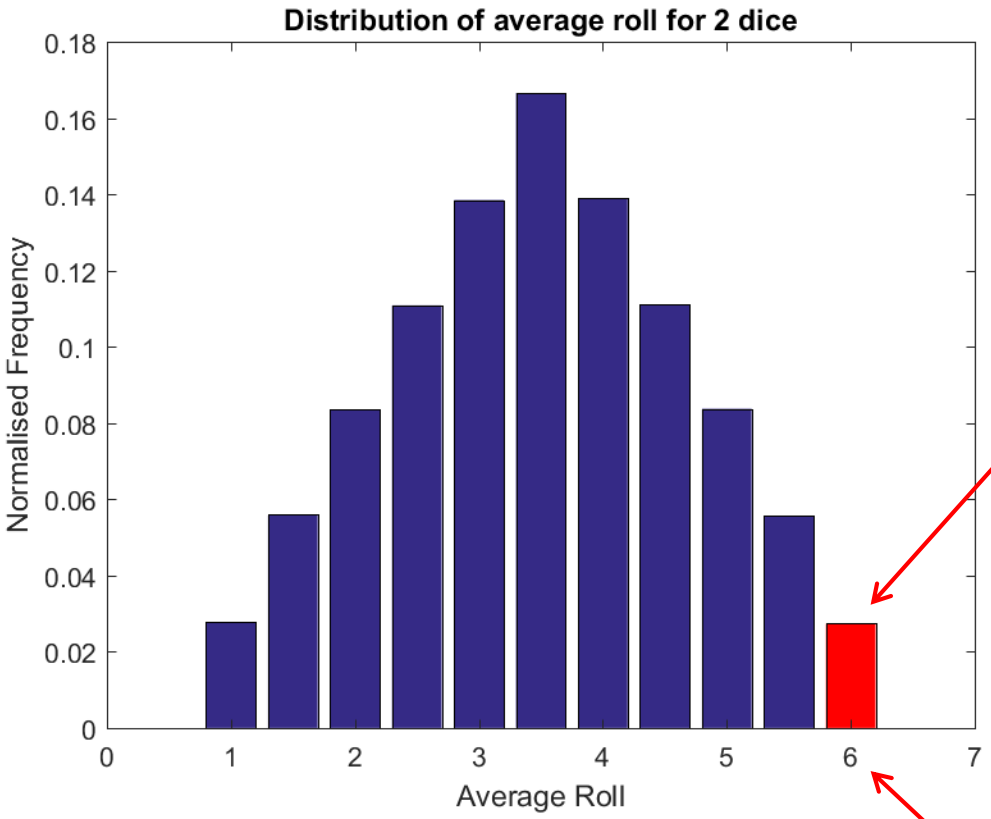
Fair Dice Distribution



Fair Dice Distribution

- What value would I need to roll (once) to reject the null (**fair dice**) hypothesis and accept the alternative (**loaded dice**) with less than 5% chance of type I error?
- Highest roll possible: 6
- Probability of rolling at least a 6: 1/6 = 0.16667
- Rejecting the null hypothesis on the roll of a 6 gives us a **16.666% chance of type I error**
  - Impossible to get sufficient evidence with a single roll of a die
- How does this change if we roll dice multiple times?

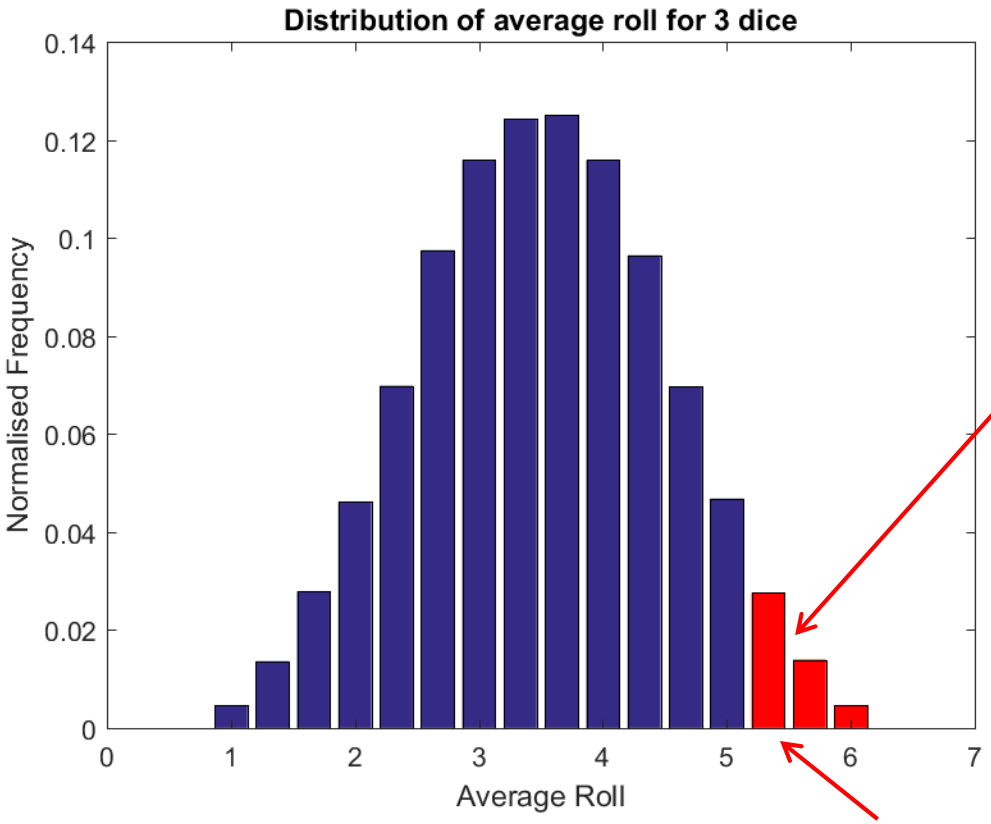
Fair Dice Distribution



5% of the distribution

Critical Value: 6

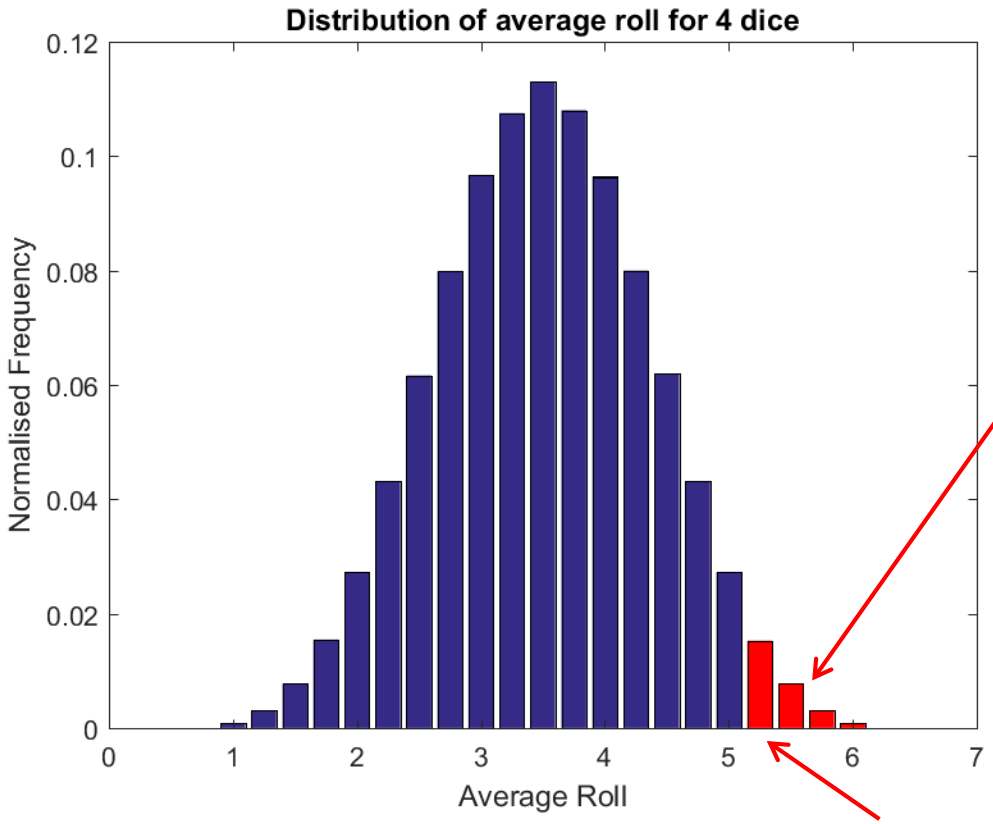
Fair Dice Distribution



5% of the distribution

Critical Value: 5.3

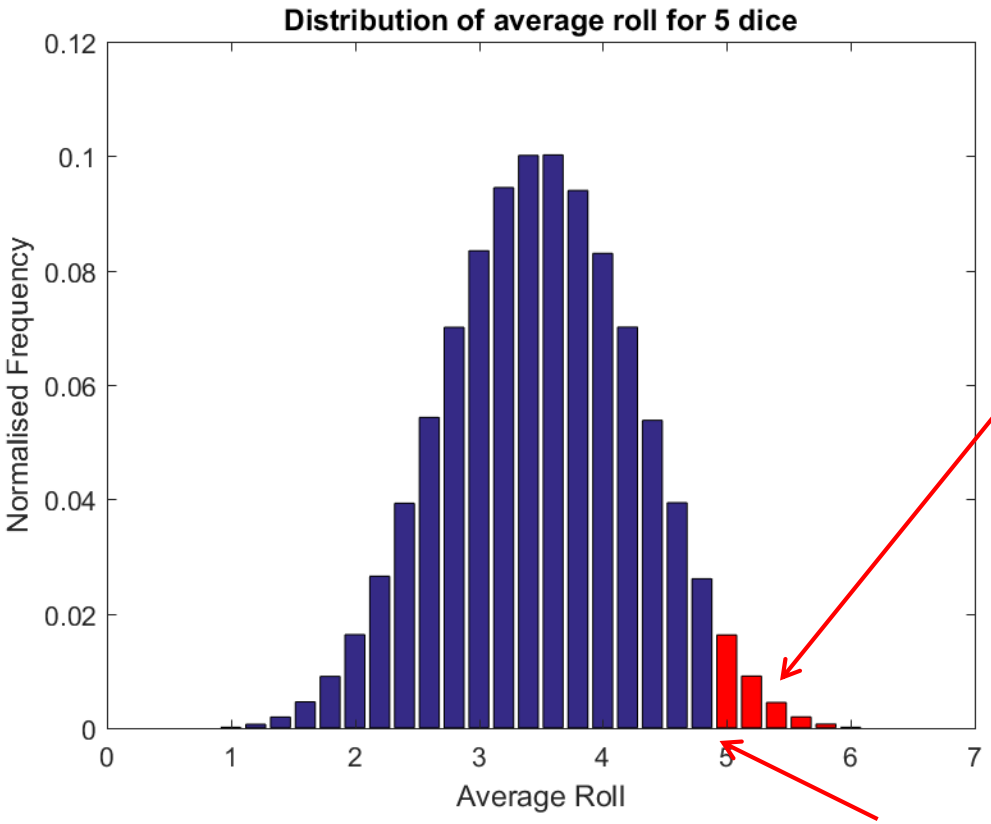
Fair Dice Distribution



5% of the distribution

Critical Value: 5.25

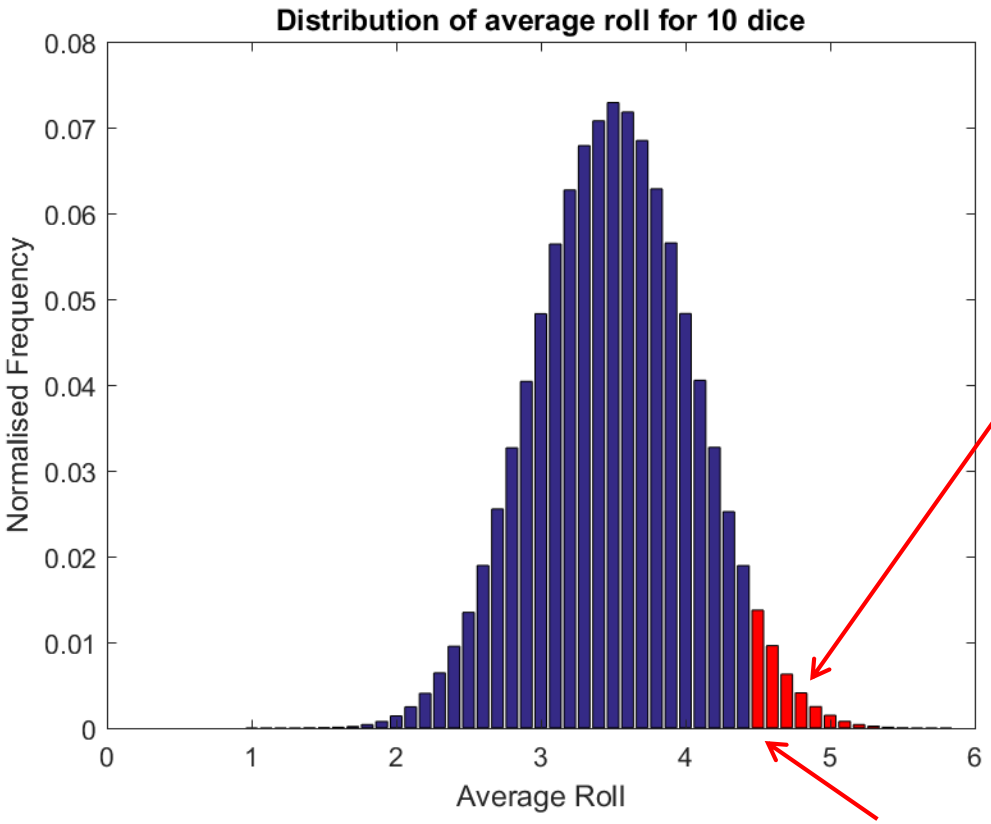
Fair Dice Distribution



5% of the distribution

Critical Value: 5

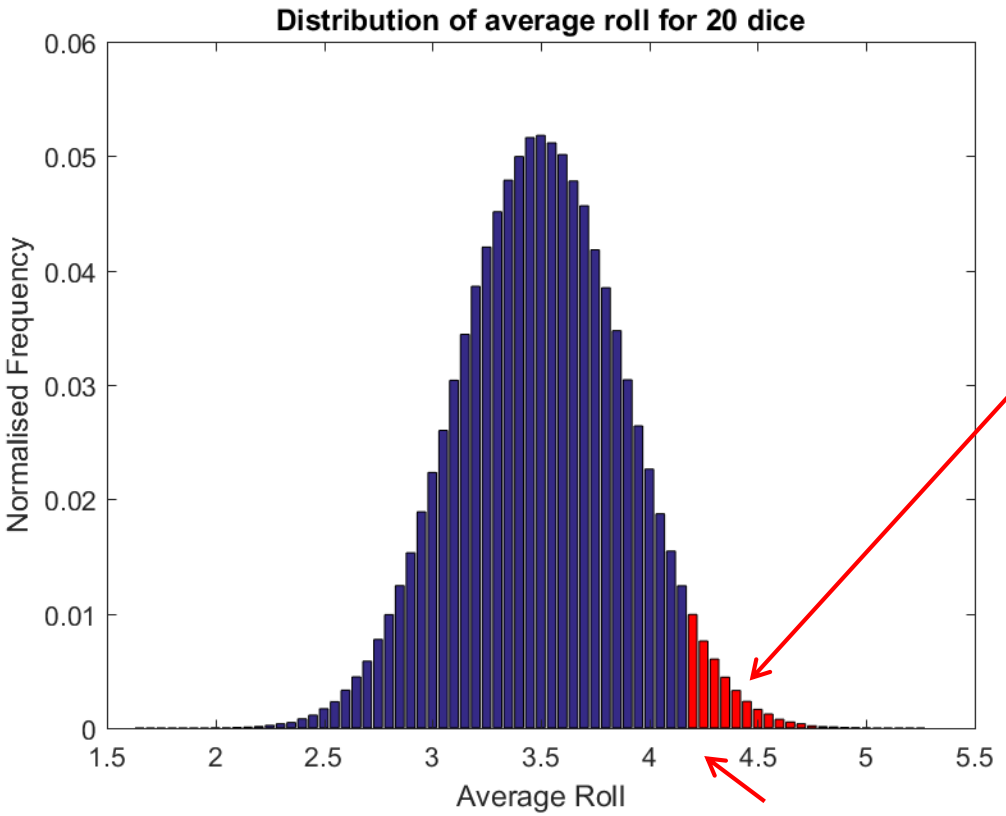
Fair Dice Distribution



5% of the distribution

Critical Value: 4.5

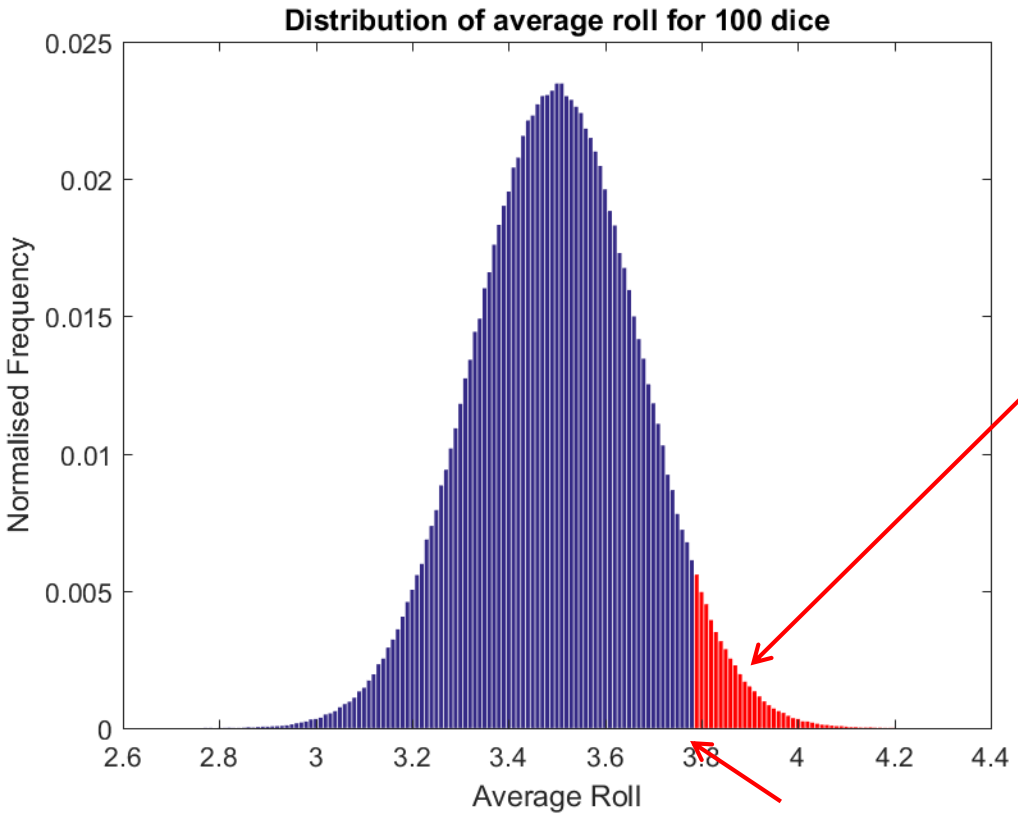
Fair Dice Distribution



5% of the distribution

Critical Value: 4.25

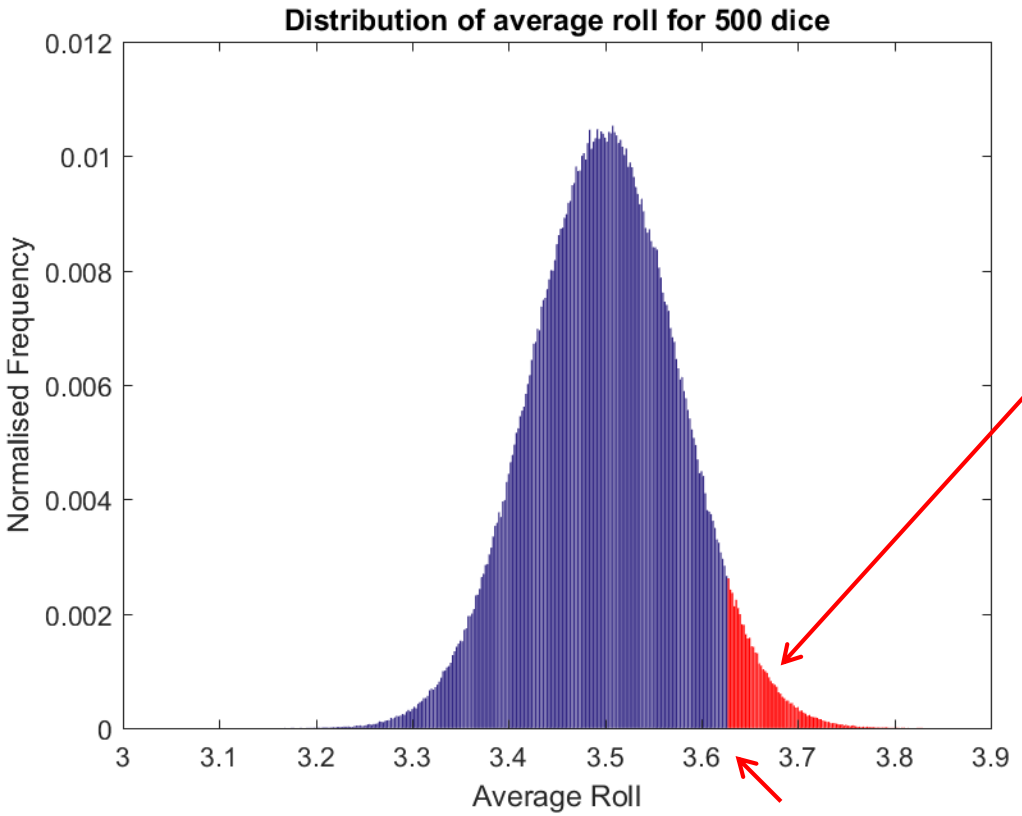
Fair Dice Distribution



5% of the distribution

Critical Value: 3.75

Fair Dice Distribution



5% of the distribution

Critical Value: 3.67

Fair Dice Distribution

Number of Dice	Critical Value
1	N/A
2	6
3	5.3
4	5.25
5	5
10	4.5
20	4.25
100	3.75
500	3.67

Caveat: these are rough estimates

Process for Estimating Distribution

- To derive these critical values, I needed a probability distribution for dice rolls
- How do I know the distribution?
- Sometimes possible to find **mathematical expression**
- For these slides, I have generated **millions of samples** and plotted a histogram
- What if samples are difficult/expensive to generate?



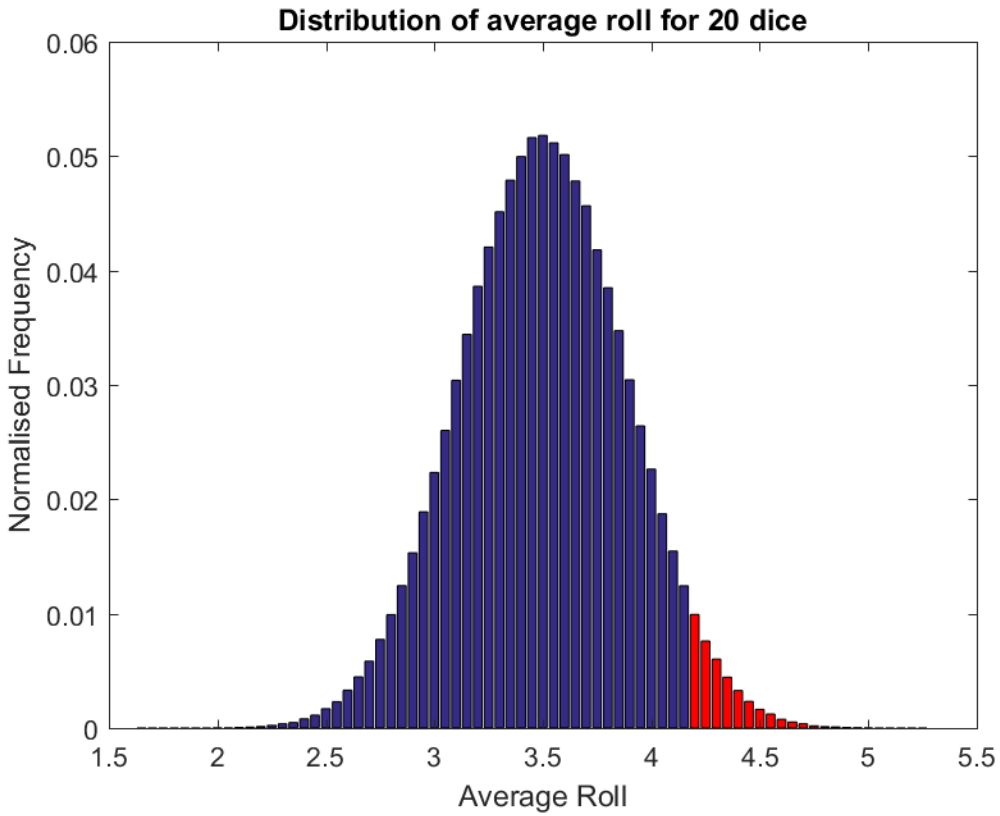
What is the **Central Limit Theorem** and what can it do for me?

Distribution of Average Rolls

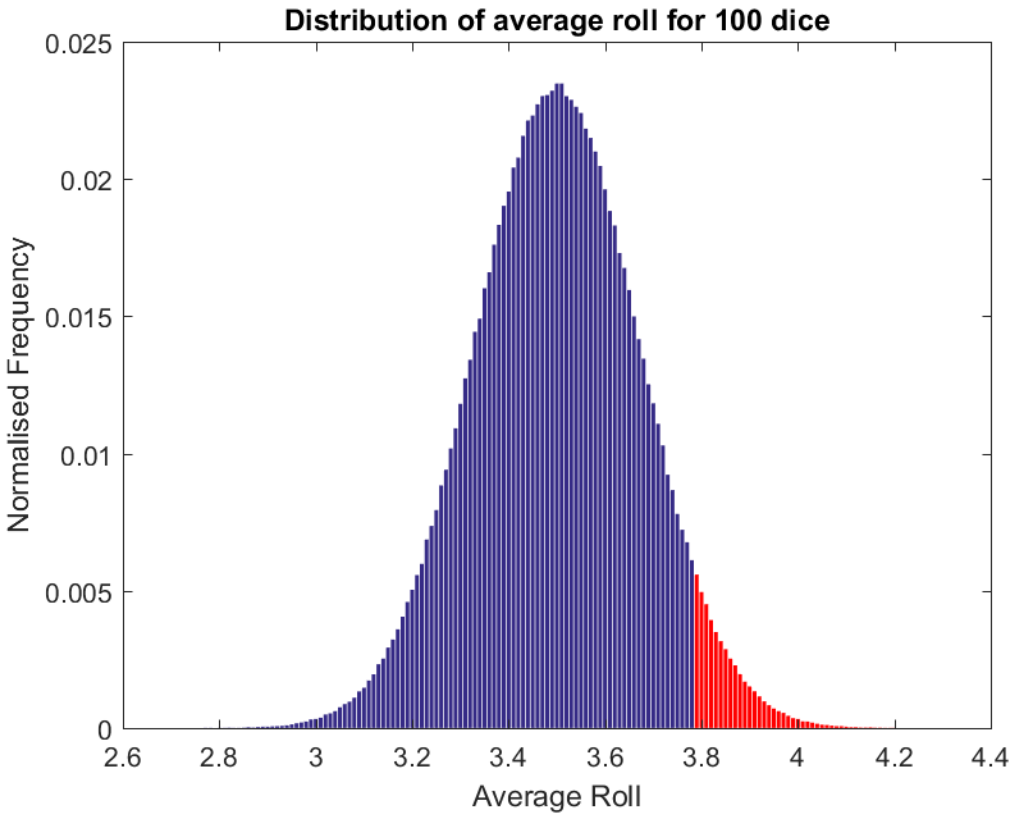
- Recall the distribution of average rolls when we were rolling large numbers of dice.



Fair Dice Distribution



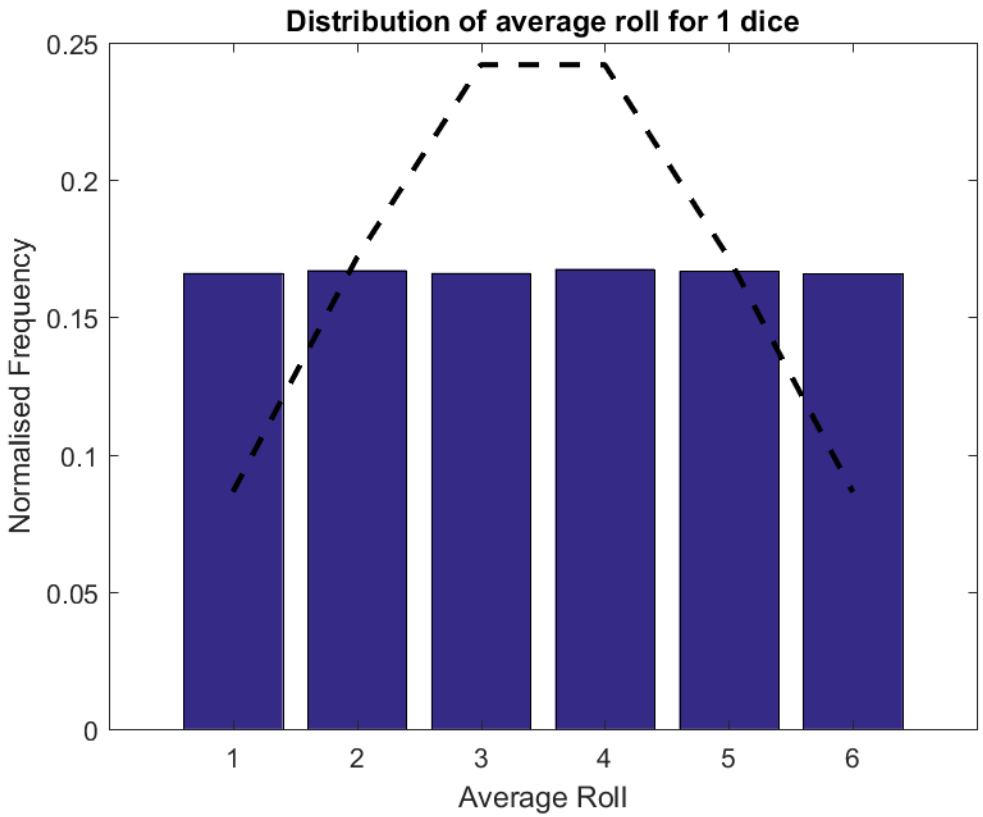
Fair Dice Distribution



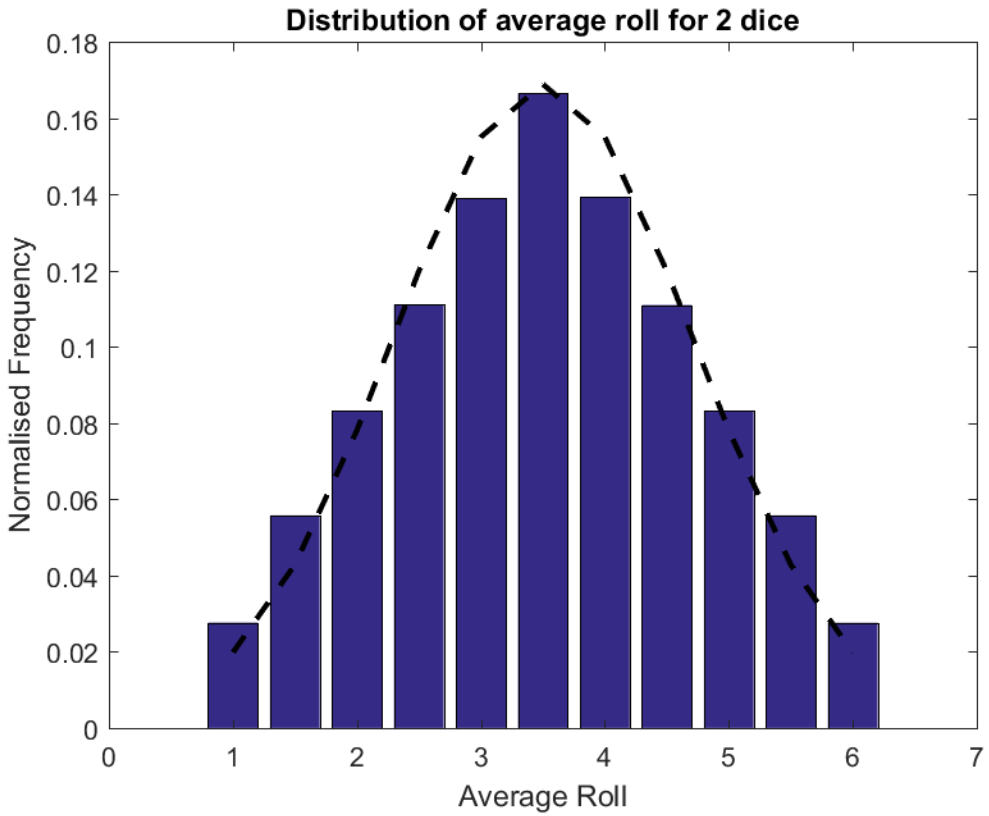
Distribution of Average Rolls

- Recall the distribution of average rolls when we were rolling large numbers of dice.
- This shape (the bell curve) is distinctive of a **normal distribution.**
- How close to a normal distribution is what we observed?

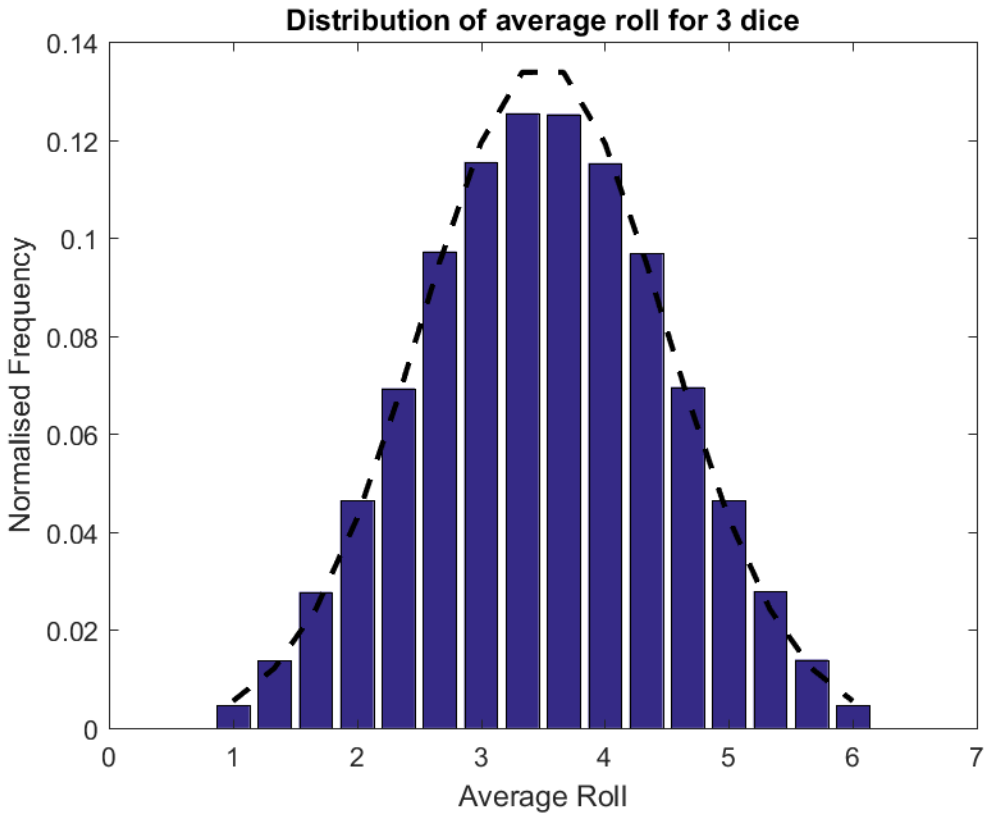
Fair Dice vs Normal Distribution



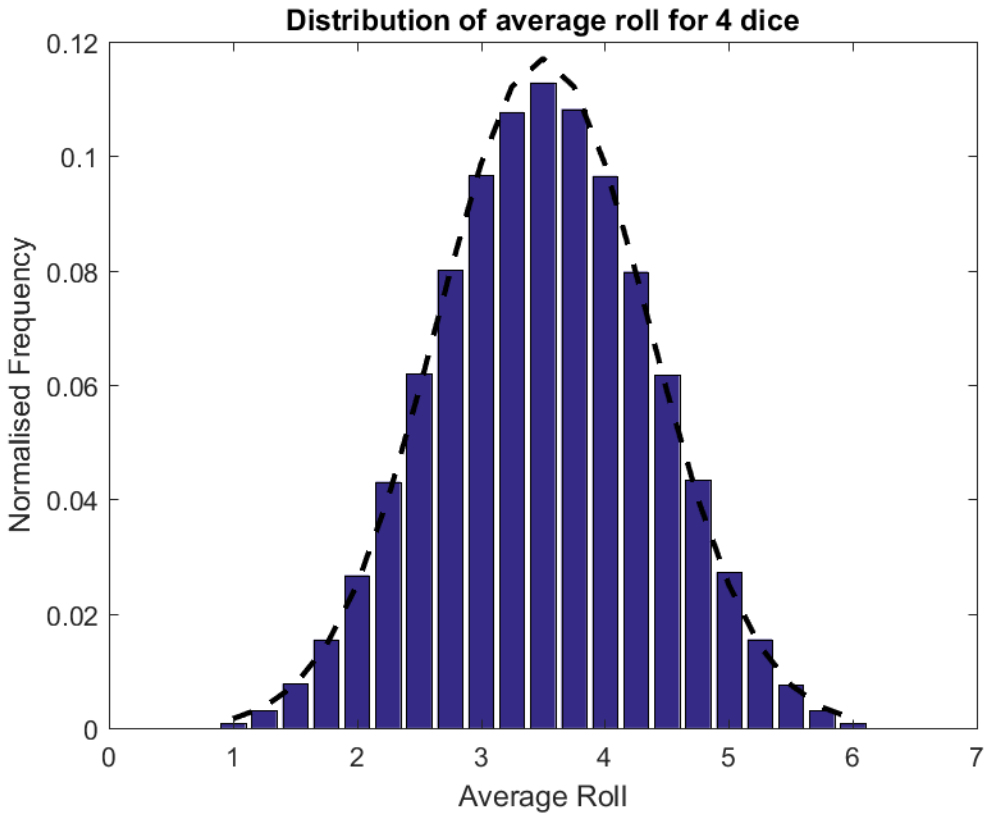
Fair Dice vs Normal Distribution



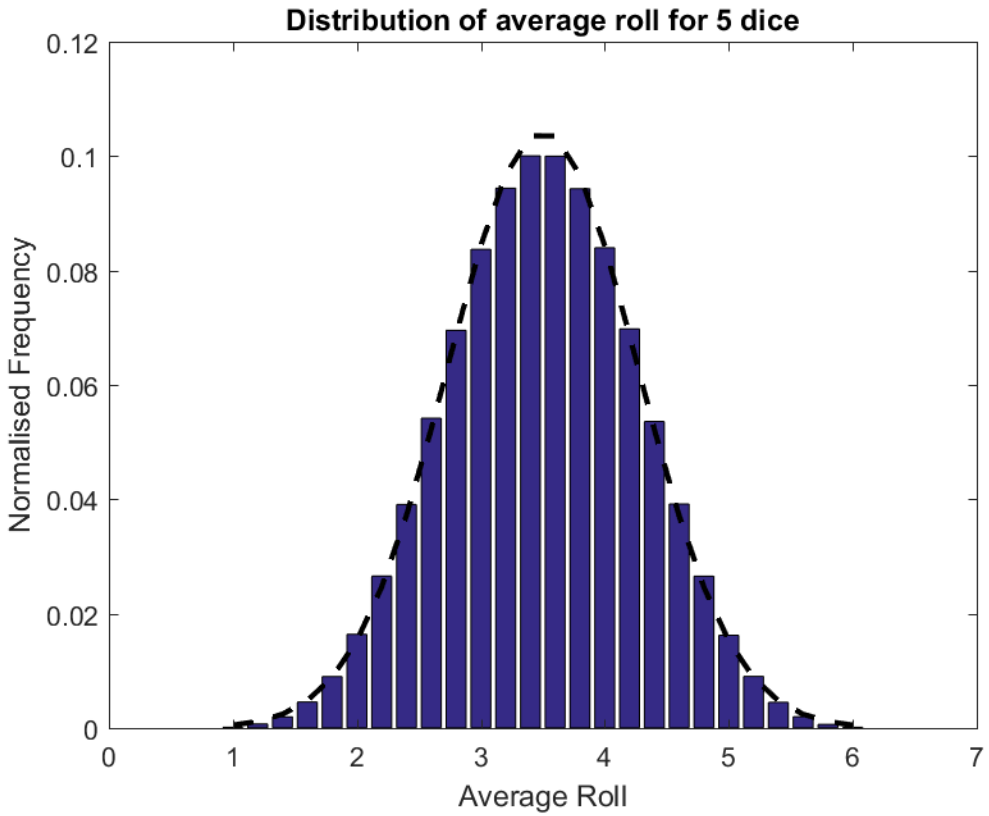
Fair Dice vs Normal Distribution



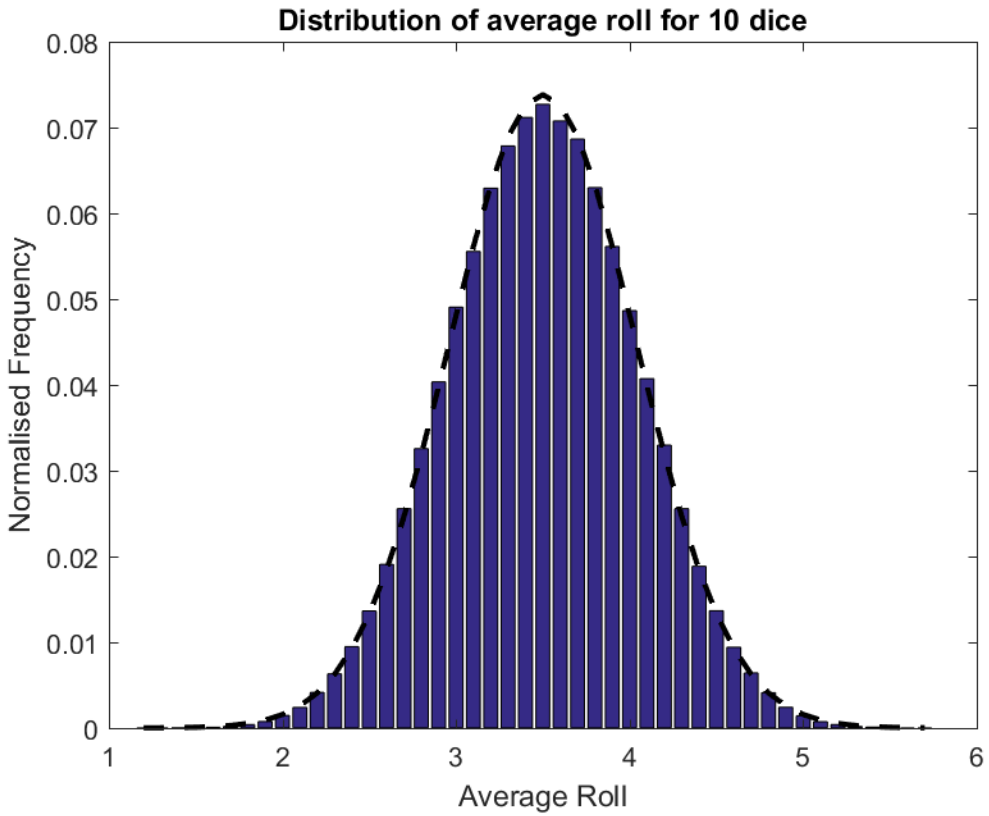
Fair Dice vs Normal Distribution



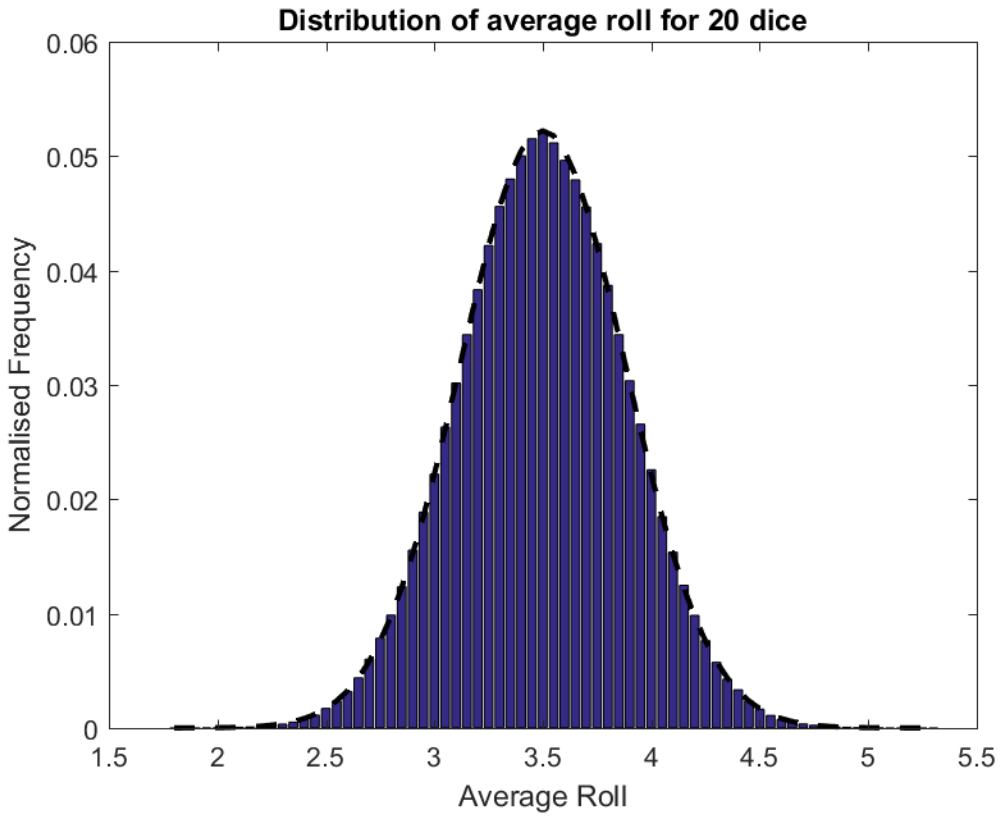
Fair Dice vs Normal Distribution



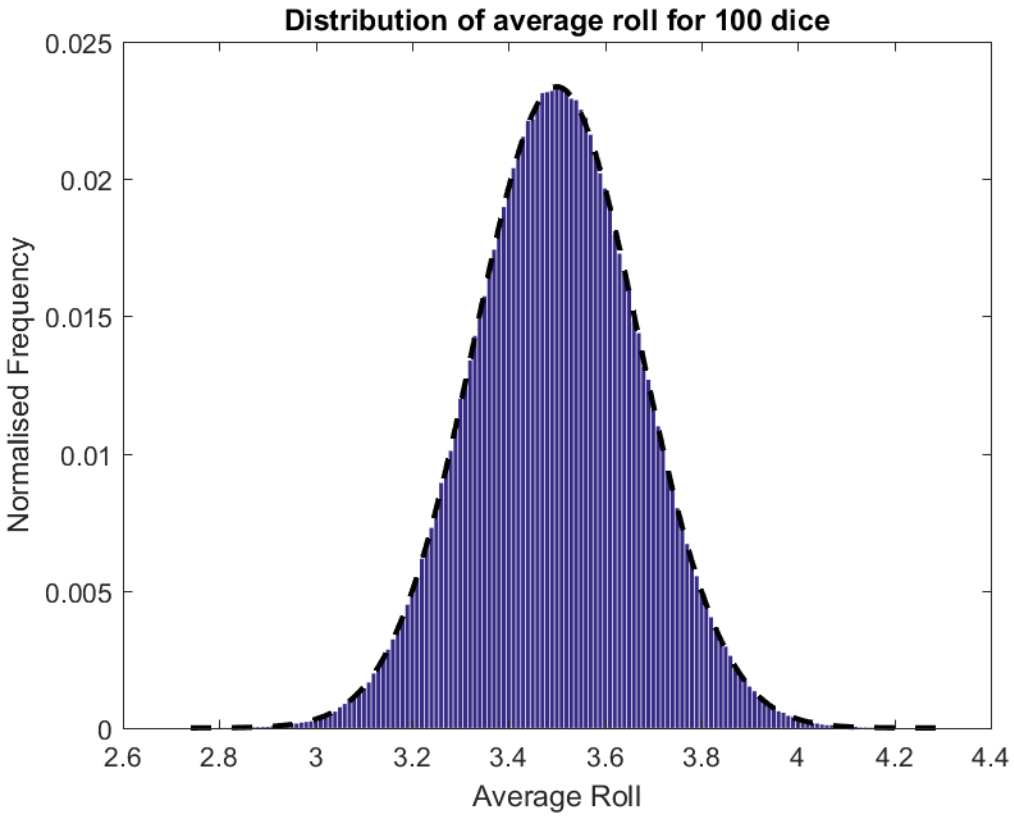
Fair Dice vs Normal Distribution



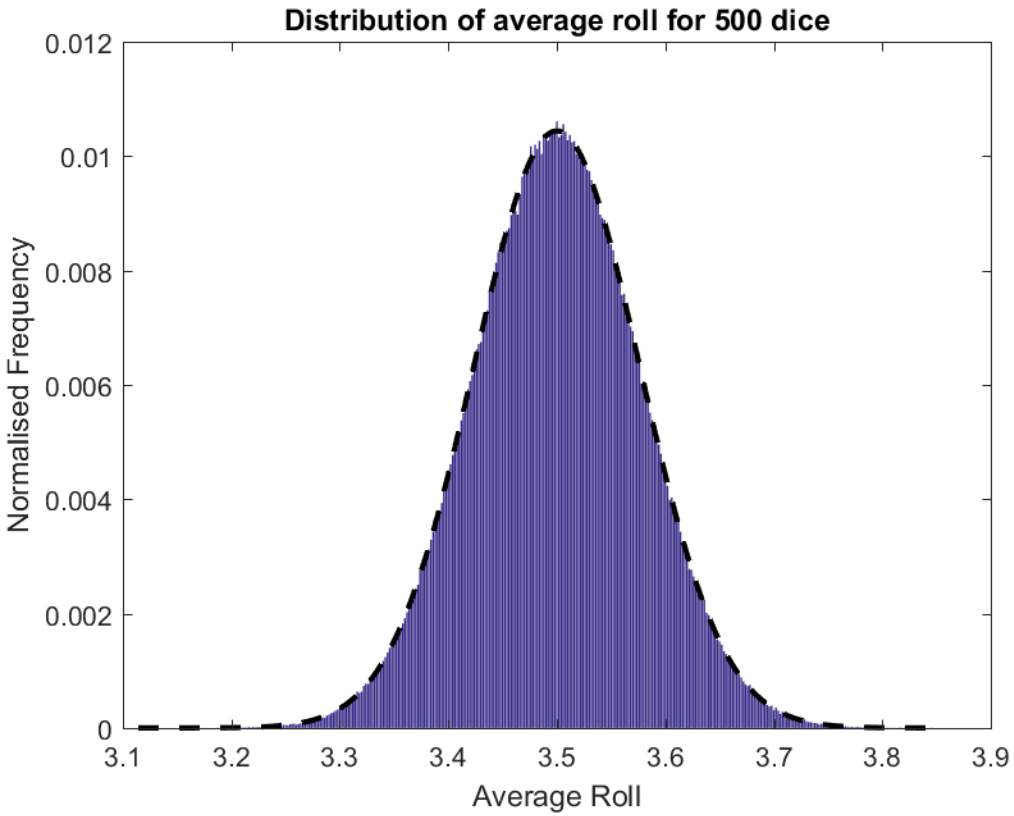
Fair Dice vs Normal Distribution



Fair Dice vs Normal Distribution



Fair Dice vs Normal Distribution



Distribution of Average Rolls

- Recall the distribution of average rolls when we were rolling large numbers of dice.
- This shape (the bell curve) is distinctive of a **normal distribution.**
- How close to a normal distribution is what we observed?
- For large numbers, the match is close to perfect!
- Is this a special property of dice?

Central Limit Theorem (1)

- Let $\{X_1, X_2, \ldots X_n\}$ be independent and identically distributed (iid) random variables with mean μ and variance σ^2.
- Let $\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$ be the sample average.
- For sufficiently large n, we can prove that $\bar{X}$:
  - **is normally distributed**
  - has **mean** μ
  - has **variance** σ^2/n.

Guideline: $n \geq 30$

Central Limit Theorem (2)

- Let $\{X_1, X_2, \ldots X_n\}$ be iid random variables with mean μ and variance σ^2.
- Let $\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$ be the sample average.
- For sufficiently large n, we can prove that $z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$:
  - **is normally distributed**
  - has **mean** 0
  - has **variance** 1.

Guideline: $n \geq 30$

Implications of Central Limit Theorem

- Why am I telling you this?
- Let us say that I have:
  - rolled 100 dice
  - observed an average roll of 3.7
- If these were fair dice, what is the probability that we would get an average result **at least this high** by chance?
- We know that, for a fair die, $\mu = \sigma^2 = 3.5$.



Implications of Central Limit Theorem

- We have:
  - $n = 100$
  - $\bar{X} = 3.7$
  - $\mu = \sigma^2 = 3.5$

- As $\mathrm{n} \geq 30$, the central limit theorem tells us that…

$$\mathrm{z} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{3.7 - 3.5}{1.871/\sqrt{100}} = 1.069$$

- …was drawn from a normal distribution with mean 0 and variance 1 **assuming that the dice are fair**

Implications of Central Limit Theorem

- We can reformulate our question…:

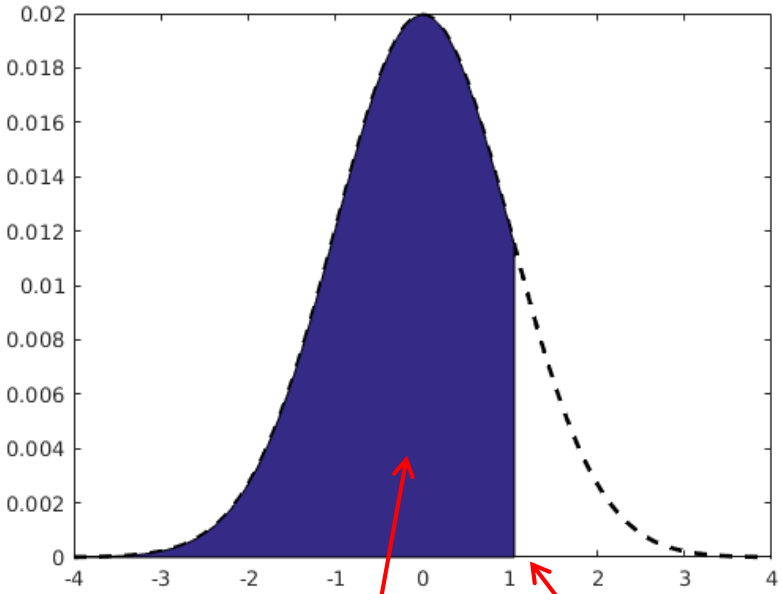
  - Let us say that I have:
    - rolled 100 dice
    - observed an average roll of 3.7
  - If these were fair dice, what is the probability that we would get an average result **at least this high** by chance?

- …as follows:

  - What is the probability of sampling a value of **at least** $z = 1.069$ by chance from a mean 0, variance 1 normal distribution?

Implications of the Central Limit Theorem

- Normal distribution is very well known
- Can look up the probabilities of getting **less than** a certain value
- For $z = 1.069$ this probability is 0.8577.
- The probability of getting **at least** $z = 1.069$ is $1 - 0.8577 = 0.1423$.



Are the Dice Loaded?

- Hopefully the connection to **hypothesis testing** is clear.

- Let's look at an example:
  - You have rolled 100 dice and observed an average roll of 3.7
  - I know that the expected average roll of fair dice is 3.5.
  - **I accuse you of using loaded dice!**
  - Use hypothesis testing to convince me that I have been too hasty with my accusations.

- Remember that you should:
  - Formulate a null and an alternative hypothesis
  - Decide whether we have enough evidence to reject the null hypothesis at significance level $\alpha = 0.05$.



Are the Dice Loaded?

- First we construct our hypotheses:
  - **Null hypothesis**: The dice are fair.
  - **Alternate hypothesis**: The dice are loaded to, on average, roll higher than fair dice.

- What is the chance of getting an average roll of at least 3.7 **assuming the null hypothesis**?
  - We have seen that it is 0.1423.
  - In English: there was a 14.23% probability that this could have happened by chance.

- Is the result **statistically significant** at $\alpha = 0.05$?
  - In English: to reject the null hypothesis we want to observe an event **extreme enough that it would happen 5% of the time, or less** assuming the null hypothesis.
  - 14.23% is greater than 5% - the result is not statistically significant!

Are the Dice Loaded?

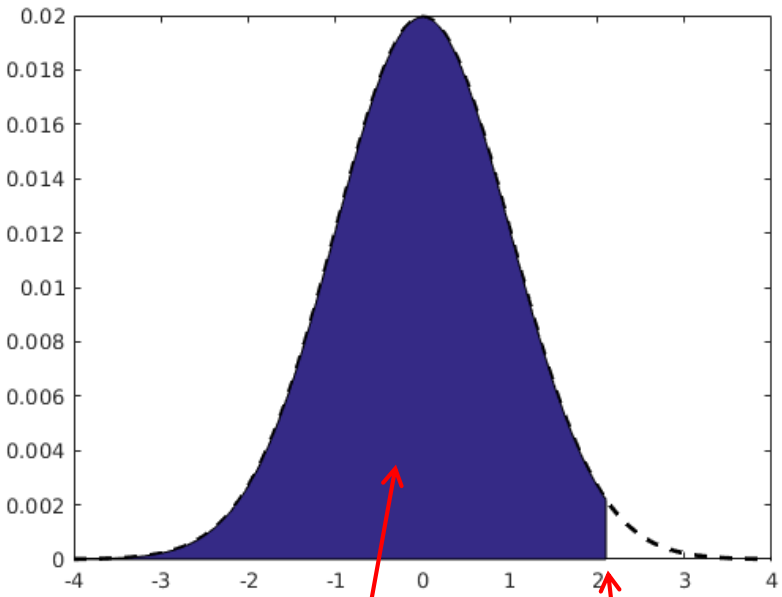
- An average roll of 3.7 from 100 dice is not statistically significant evidence that the dice are loaded (at $\alpha = 0.05$).

- Conclusion:
  - ~~We have enough evidence to say that the dice are not loaded~~.
  - We **do not have enough evidence** to say that the dice are loaded.



Second (Quick) Example

- I have rolled 100 dice, observing an average roll of 3.9. Can I conclude that the dice are loaded?
- This gives us $z = 2.1381$
- The probability of getting a result at least this extreme is $1 - 0.9838 = 0.0162$ **p-value**
- 1.62% is less than 5% - the result is statistically significant!



How can I apply these ideas to validating claims about algorithms?



Practical Obstacles

- If we want to compare two algorithms on a problem, we are unlikely to know the **true mean and variance** of either.
- We need to look at the difference between two distributions.
- The difference between two normal distributions is also normal (**good**)…
- …but estimating the mean and variance biases the test (**bad**).
- Standard approach: use ***Welch's t-test***.
  - Exactly the same idea as the normal distribution method, but controls for biases.



Welch's t-test

For the two sample averages $\bar{X}_1$ and $\bar{X}_2$ with variances σ_1^2 and σ_2^2 (assumed unequal) the t-statistic is calculated

- $t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\sigma_1^2 + \sigma_2^2 / N}}$

- The null hypothesis -- that the two averages are equal -- can be rejected …

  ….if $p < \alpha$

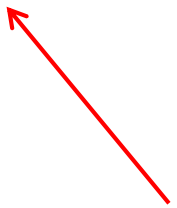
Experimental Design

- I have:
  - two algorithms, **A** and **B**,
  - a problem, **P**,
  - a metric, **M**.
- I think that **A** should outperform **B** on problem **P** in terms of metric **M**.
- How do I set up an experiment to try and gather evidence for this?



Experimental Design

- **Step 1**: Set up null and alternate hypotheses
  - **Null hypothesis**: *Algorithm A is, on average, no better than algorithm B on problem P in terms of metric M.*
  - **Alternate hypothesis**: *Algorithm A is, on average, better than algorithm B on problem P in terms of metric M.*



Statement that we want to prove

Experimental Design

- **Step 1**: Set up null and alternate hypotheses
- **Step 2**: Test algorithms A and B on problem P **at least 30 times**. Record the results.
  - If we think that the difference will be small, we can run more trials.
- **Step 3**: Apply Welch's t-test to get the **p-value.**
- **Step 4**: If the p-value is less than our significance level ($\alpha = 0.05$), we can reject the null hypothesis.



How can I avoid invalidating my claims?



Pitfalls

- The method we have discussed is precisely calibrated to justify our conclusions.
- Very easy to modify to make the claims invalid – be careful!
- Are there any problems in the following scenarios?



Scenario 1

- I have:
  - two algorithms, **A** and **B**,
  - a problem, **P**,
  - a metric, **M**.

- I test **A** and **B** 30 times on problem **P.**
- I see that **A** outperforms **B** in terms of metric **M.**
- I use this same data to test the hypothesis that **A** is better than **B** (for the given problem and metric).
- Welch's t-test gives me a p-value of 0.04, so I reject the null hypothesis at significance level ($\alpha = 0.05$).



Scenario 2

- I have:
  - an algorithm, **A**,
  - a problem, **P**, which is representative of some broader class **Q**,
  - a metric, **M**.

- I think that I can design an algorithm better than **A** on class **Q.**
- I design an algorithm **B** and refine it by testing it on problem **P.**
- I test the hypothesis that **B** is better than **A** on problem **P** and find sufficient evidence at significance level $\alpha = 0.05$.
- As **P** is representative of class **Q**, I claim that **B** is better than **A** at solving problems in class **Q**.



Summary

- At the end of this lecture you should…
  - …know which types of comparative claims we can reasonably make about the performance of optimisation algorithms,
  - …be able to formulate these claims such that they are testable,
  - …design an experiment to test your claims.

